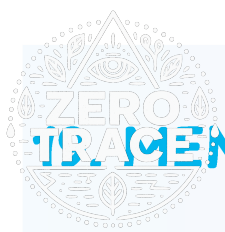




Q. A 45-year-old male from a village presents to the ER with a sudden onset of fever and altered mental status. Neurological exam reveals GCS: 9/15 and (+) bilateral Babinski sign. Work up shows HB- 8gm, TLC- 4000 cell/cubic mm, elevated LDH with elevated S. creatinine and S. bilirubin. Which of the following diagnostic tests should be prioritized for this patient?

- a. HRP-2 card test
- b. PCR for HSV-1 on CSF
- c. MRI brain
- d. EEG with sleep deprivation

Ans: (a)



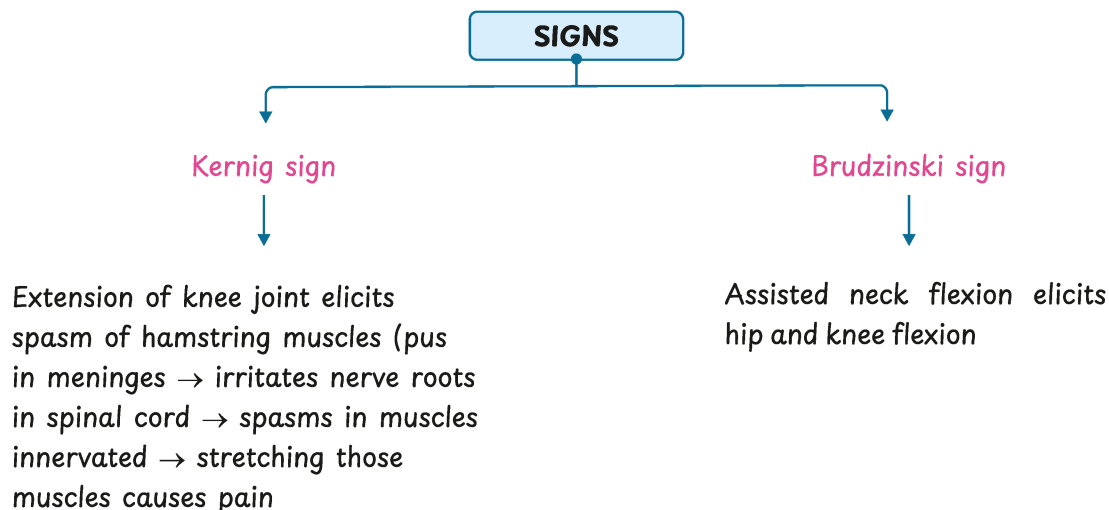
MENINGITIS PART 2

FEATURES

- Unable to follow commands
- Nuchal rigidity (+)

SIGNS OF MENINGEAL IRRITATION

00:01:58



WORK-UP FOR SUSPECTED PYOGENIC MENINGITIS

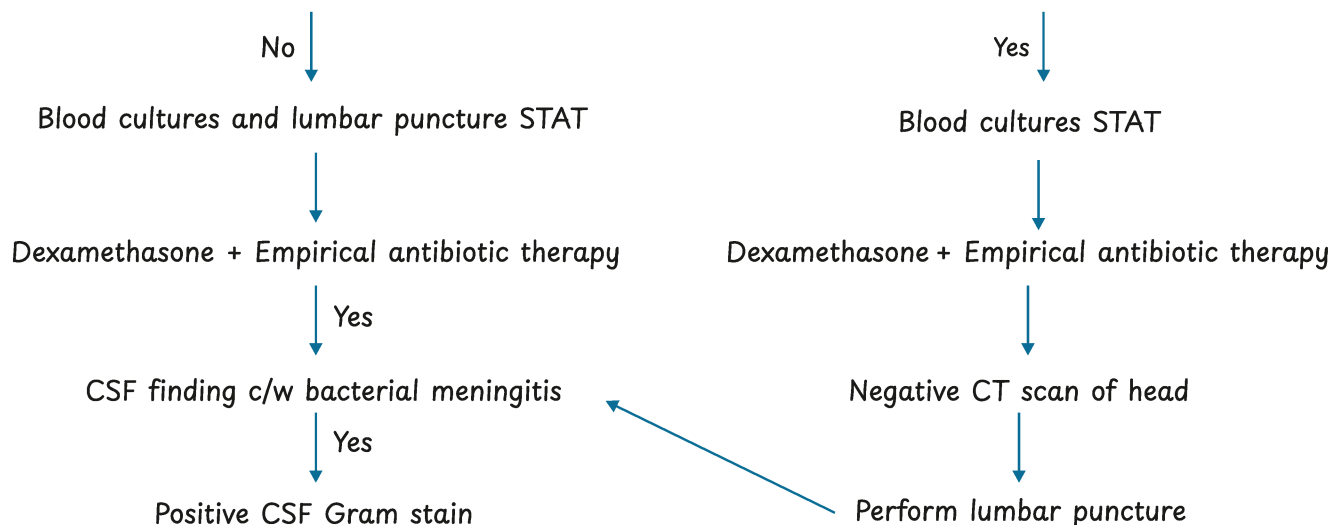
00:03:25

- Blood culture
 - Neuroimaging/Lumbar puncture
 - Focal neurological deficit
 - Focal seizures
 - Fundus - presence of papilledema
 - ↑ ICP, Immunocompromised
 - Trauma
 - Space occupying lesion
 - Guarded lumbar puncture (to avoid foraminal herniation)
 - CSF - turbid
 - M/E - PMN cells, Gram stain done
 - Empirical antibiotics started within < 60 mins of admission
- Neuroimaging (Lumbar puncture - C/I)



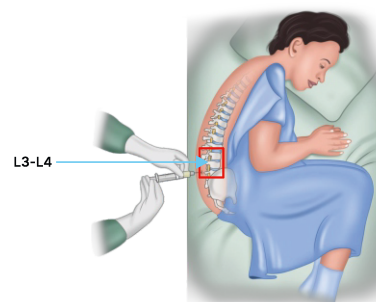
Suspicion for bacterial meningitis

Immunocompromised, h/o CNS disease, new onset seizure, papilledema, altered consciousness, focal neurologic deficit, delay in performance of diagnostic lumbar puncture

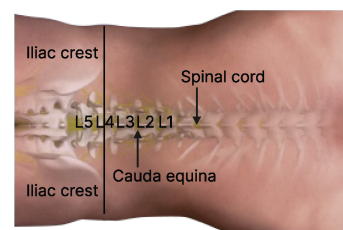


LUMBAR PUNCTURE

- **Site** of lumbar puncture
 - L3 - L4 > L4 - L5
- **Position** of patient
 - Lateral decubitus position
 - Knee chest position (to get curvature of spine)
- **Anatomical landmarks**
 - Identify Iliac crest
 - **Tuffier line** - an imaginary line connecting both sides of Iliac crest
→ It intersects spinous process of L4 vertebra
- **Site preparation**
 - Aseptically cleaned with **Chlorhexidine**



00:12:34



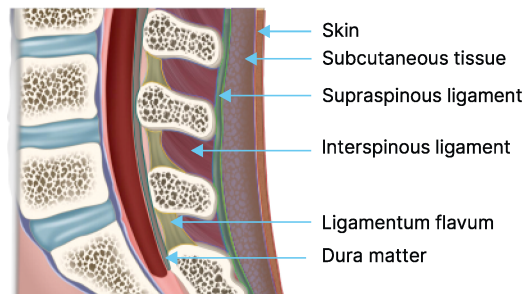
C/I TO LP



- Administration of local anesthesia
- **Procedure**
 - Needle introduced with bevel facing upwards
 - Progress needle in **cephalad** direction (15°)
 - Aim for umbilicus
- Layers pierced during LP

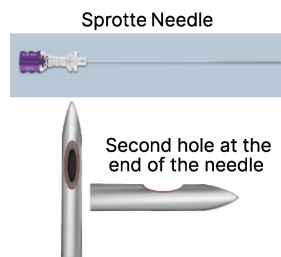
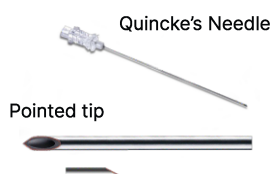


- Skin
- Subcutaneous tissue
- Supraspinous ligament
- Interspinous ligament
- Ligamentum flavum
- Dura mater
- Arachnoid layer



TYPES OF LUMBAR PUNCTURE NEEDLES

00:23:00



QUINCKE'S NEEDLE	SPROTTE NEEDLE
- Needle has double-edged bevel, needle goes into dura mater to separate fibers - When needle is taken out, it will evert edges → leakages → headaches	- Bevel is on side - Atraumatic needle, minimized complications

POST LP HEADACHE

- Dura mater breached during LP
- Fibers in dura mater split during process
- Leakage of CSF

LOW VOLUME BLOOD PATCH

- Post LP headache not resolved to conventional T/t measures, **blood is injected up to dura mater** → into epidural space → to create seal over punctured Dura mater

STEPS IN MANAGEMENT OF POST LP HEADACHE

00:27:00

- Strict bed rest
 - 8-10 hours supine position rest
- Analgesics if required - COX-1 inhibitors, diclofenac sodium
- DOC - I/V caffeine**
- Last resort - Opioids (Codeine)
 - S/E - constipation**
- Laxatives, cough suppressants
- Low volume blood patch



CSF SAFELY REMOVED

Diagnostic purposes
- 20 ml

Therapeutic purposes
- 30 ml infection

LP IS DONE FOR THERAPEUTIC RELIEF IN -

- Normal pressure hydrocephalus (mnemonic - wet, whacky, wobbly)
- Removal of CSF → Fisher test → improves gait abnormality of patient
 - Urinary incontinence (d/t dilated ventricles pressing against paracentral lobule)
 - Dementia (d/t pressure exerted to grey matter against calvarium)
 - Gait abnormalities (M/c - magnetic gait) (d/t pressure over frontal lobule)

NUMBER OF TUBES USED TO COLLECT CSF

TUBES

Sugar and protein analysis

Cell count

Culture and Gram stain

Special studies - CSF ELISA for Cryptococcal antigen, India ink, CSF PCR for MTB infection

- Minimum - 3 (for special studies - 4 test tubes)

EVALUATION OF CSF

00:39:33

	OPENING PRESSURE	CELLS	SUGAR	PROTEIN	GROSS APPEARANCE
NORMAL	50-180 mmH ₂ O	0-5 lymphocytes	> 0.6 (> 2/3 of blood sugar)	15-45 mg/dl	Clear
ACUTE BACTERIAL MENINGITIS	↑	>1000 PMN cells	< 0.6	↑ (Antibodies)	Turbid/ cloudy
TUBERCULOUS MENINGITIS (PEM Grade IV, No BCG scar, Contact with Koch's)	↑	>100 lymphocytes	< 0.6	↑↑ (Cobweb coagulum)	Cloudy
VIRAL MENINGITIS (short history)	↑	50-500 lymphocytes	Normal	↑	Clear/ turbid
SUBARACHNOID HEMORRHAGE	↑	RBC ++ (Xanthochromic CSF)	Normal	Normal	Bloody

GCS (Lymphocyte demyelination) Cytobuminological dissociation	Normal	Normal	Normal	↑ (Antibodies)	Clear
---	--------	--------	--------	----------------	-------

- CNS leukemia** - Presence of malignant cells in sediment of CSF
- Fungal meningitis** - lymphocytosis, eosinophilia, hypoglycorrhachia (low CSF glucose), ↑ protein, send sample for special studies of CSF

TUBERCULAR MENINGITIS

00:54:38

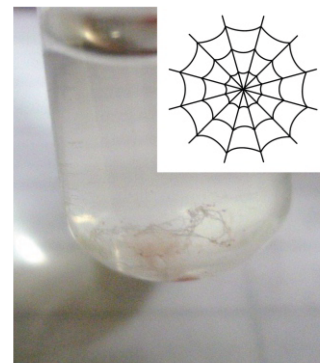
Case - 5 year old PEM grade III child is brought to ER with complaints of low grade fever for last one month. Mother says child is very irritable and keep vomiting and has minimal oral intake. For last 24 hours child is not opening eyes and sleeping very deeply. O/E: GCS- 9/15, Kernig sign and Brudzinski sign are noted with bilateral extensor plantars. Fundus shows papilledema. CXR show left sided hilar lymphadenopathy

Ans - Inference from question -

- Infective cause
- Altered sensorium
- (+) signs of meningeal irritation (Kernig and Brudzinski sign)
- Raised ICP
- Involvement of corticospinal pathway

WORKUP OF SUSPECTED TB MENINGITIS

- NCCT head
 - ↑ ICP, hydrocephalus, basal exudates
- Guarded LP** after giving mannitol and control of raised ICP
 - Cobweb coagulum
 - M/E - 1000 lymphocytes/ cu mm (lymphocytic pleocytosis)
 - Sugar ↓, Protein ↑↑ (gm/dl)
- Mantoux - false negative
- CXR and USG abdomen
 - Disseminated TB (hepatosplenomegaly)
- HIV status

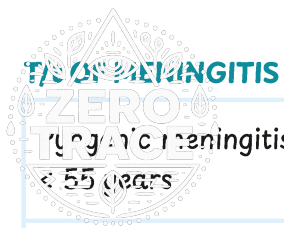


T/t OF TUBERCULAR MENINGITIS

01:00:48

- ATT** (12 months for children, 6 months for adults)
- Steroids
 - Dexamethasone - 6 weeks**
 - Tapering dose (to decrease vasogenic cerebral edema)
- Ventriculoperitoneal shunting
 - Evidence of hydrocephalus
- Steroids** used in following conditions
 - TBM
 - NCC
 - Brain tumor

} Vasogenic cerebral edema +



TUBERCULOSIS

01:02:32

Pyogenic meningitis in patient aged > 3 months and < 55 years	Ceftriaxone + Vancomycin
Tubercular meningitis	ATT + Steroids
Cryptococcal meningitis	Liposomal Amphotericin B + 5-flucytosine
Cerebral malaria	Artesunate



Q. Which of the following is not useful in management of post dural puncture headache?

- a. Caffeinated drinks
- b. Plenty of I/V fluids
- c. Blood patch
- d. Laxatives

Ans: (b)

Q. A 5 year old PEM grade II child presents to ER with fever, headache, vomiting episodes past 3 days and focal seizure episodes on arrival to hospital. On examination meningeal signs are noted. Fundus shows papilledema. Which of the following investigations is not to be done in this patient?

- a. Lumbar puncture
- b. Neuroimaging of CNS
- c. Complete blood counts and blood culture
- d. Chest X ray

Ans: (a)

Q. A 70 year old patient presents with progressive gait disturbance, urinary incontinence and cognitive decline. Neurologist makes a diagnosis of normal pressure hydrocephalus. Which of the following will be done to reduce gait disturbances in this patient?

- a. Lumbar puncture with removal of 20-30 ml of CSF
- b. Osmotherapy with Mannitol
- c. Ventriculostomy
- d. Ventriculo caval shunt

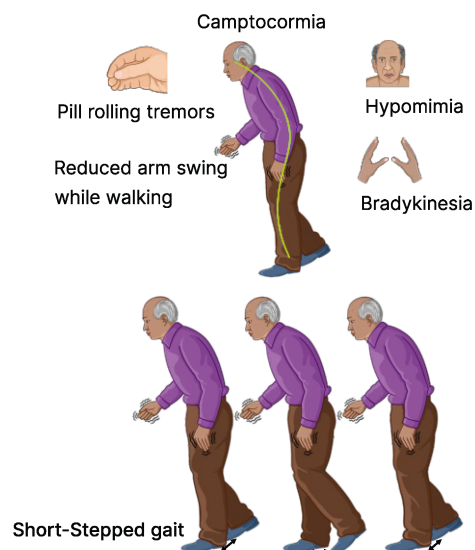
Ans: (a)



12. PARKINSONISM & PARKINSONS DISEASE



- M/c cause of Parkinsonism → **Parkinson's disease**
- Athletes in body contact sports have ↑ incidence of neurologic disorders
- **Micrographia** (small handwriting): d/t rigidity of hand muscles
- Earliest motor manifestation: **resting tremor**
- ↓ arm swing while walking
- **Camptocormia** (stooped posture while walking)
- Must have criteria - **Bradykinesia**
- **Hypomimia** → akinetic mutism, expressionless face
- **Pill rolling tremor** (between the thumb and index finger) (low frequency tremor)
- **Short-stepped gait** (higher tendency to fall)



BASIC POINTERS

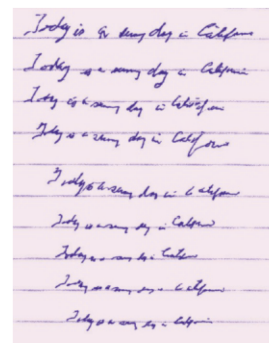
	PARKINSON DISEASE	ALZHEIMER'S DISEASE
Neurotransmitter	Dopamine ↓	• Acetylcholine ↓ • Norepinephrine ↓ • Serotonin ↓
Area involved	Substantia nigra	Basal nucleus of Meynert
Bodies seen	Lewy bodies (α-synuclein protein)	Hirano bodies (Tau protein)

Important Information

- Leading cause of **cortical dementia**: **Alzheimer's disease**
- Leading cause of **subcortical dementia**: **Parkinson's disease**

BRAAK STAGING: PROGRESSION OF LEWY BODY PATHOLOGY

- Peripheral enteric system :**Constipation** (non-motor manifestation)
- Anterior olfactory nucleus :**Anosmia** (non-motor manifestation)
- Dorsal motor nucleus of Vagus :Cardiac denervation → **orthostatic hypotension**
- **Substantia nigra pars compacta**: Hallmark of PD
- Cerebral hemispheres : Cognition issues



Micrographia

00:06:13

BASAL GANGLIA FUNCTION

00:10:01

- Facilitation of selected movements
- Suppression of unwanted movements

OUTPUT REGION OF BASAL GANGLIA

- Normal output: globus pallidus interna + substantia nigra → thalamus → motor area
- Globus pallidus have inhibitory control over the thalamus
- **Dopamine deficiency**: ↑ output from inhibitory area → ↑↑ inhibition in thalamus → bradykinesia

CARDINAL FEATURES

00:12:04

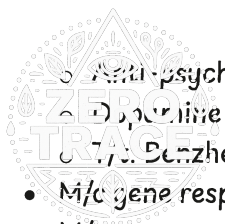
Mnemonic: **TRAP**

- **T**remors
 - Resting tremors → 4-6 Hz
 - Asymmetrical
 - **Pill rolling tremor**
- **R**igidity
 - **Cogwheel** rigidity (upper extremity)
 - **Lead pipe** rigidity (legs)
- **B**radykinesia
 - Contributes to disability
 - Akinetic mutism
 - Must have criteria for PD
- **P**ostural instability
 - Falls
 - **Camptocormia** (stooped posture): abnormal thoracolumbar flexion (45°)
→ Lie in bed: straight
 - **Festinating gait**

NON MOTOR FEATURES	OTHER MOTOR FEATURES
• Anosmia • Autonomic dysfunction ◦ Orthostatic hypotension (SBP- ↓20mmHg; DBP- ↓10 mmHg) ◦ Neurogenic bladder • REM behavior disorder • Cognition impairment	• Drooling of saliva • Eye blinking ↓ • Falls, freezing (↓initiation of a movement) • Hesitancy, hypomimia (masked-like facies), hypophonia • Micrographia

One liners:

- 2nd M/c cause of age related neurodegeneration
- M/c cause of parkinsonism:
 - Parkinson disease
 - T/t: Levodopa, carbidopa
- M/c cause of secondary parkinsonism:
 - **Drug induced parkinsonism**



- Anti-psychotics: Haloperidol, chlorpromazine
- Dopamine ↓, ↑ acetylcholine
- T/A: Benzhexol, Trihexiphenidyl
- M/c gene responsible for PD: **GBA** (encodes for lysosomal glucocerebrosidase)
- M/c gene responsible for Juvenile onset PD (< 20 years): **PARKIN gene**
- M/c gene involved in sporadic PD: **LRRK2**

DRUG INDUCED PARKINSONISM

00:28:06

- Haloperidol, Chlorpromazine
- Metoclopramide
- Tetrabenazine (dopamine depleter)
- Cinnarizine (Anti-vertigo drug)
- Flunarizine
- Amiodarone
- Lithium

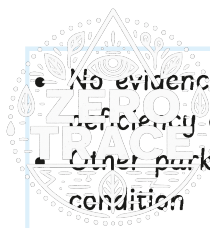
Important Information

- Neurological disorder with excess of Dopamine: Huntington chorea

CRITERIA FOR DIAGNOSIS OF TYPICAL PARKINSONISM

UK BRAIN BANK CRITERIA	MOVEMENT DISORDERS SOCIETY CRITERIA
• Response to Levodopa • Asymmetrical tremors	• At least 2 supportive criteria • No red flags (cerebellar features) • Absence absolute criteria • Supportive criteria ◦ Dramatic response to Levodopa ◦ Peak dose dyskinesia ◦ Asymmetrical resting tremors ◦ Anosmia and cardiac denervation with FD-PET

ABSOLUTE EXCLUSION CRITERIA	RED FLAGS	SUPPORTIVE CRITERIA
• Cerebellar signs • Supranuclear gaze palsy • Established diagnosis of BVFTD • Parkinsonism restricted to the lower limbs only for >3 years • Treatment with an antidopaminergic, or with dopamine-depletion agents • Absence of response to levodopa • Sensory-cortical loss	• Rapid deterioration of gait • Absence of motor symptom progression over 5 years • Early bulbar dysfunction (CN palsy) • Respiratory dysfunction • Early severe autonomic failure • Early recurrent falls due to a misbalance • Disproportionate anterocollis	• A clear and dramatic positive response to dopaminergic therapy • Levodopa-induced dyskinesia • Documentation of resting tremor of a limb • A positive diagnostic test of either olfactory loss or cardiac sympathetic denervation on scintigraphy



- | | |
|--|---|
| <ul style="list-style-type: none"> • No evidence for dopaminergic deficiency on functional imaging • Other parkinsonism-inducing condition | <ul style="list-style-type: none"> • Absence of common non-motor features of disease during >5 years • Pyramidal tract signs • Bilateral symmetric presentation |
|--|---|

ATYPICAL PD (PCMF)	SECONDARY PARKINSONISM (DHL-VIT)
• Progressive supranuclear gaze palsy ◦ Involvement of superior colliculus ◦ Function: downward gaze • Corticobasal syndrome (alien limb phenomenon) • Multisystem atrophy • Frontotemporal dementia (anti-social behavior)	• Drugs: Anti-psychotics • Normal pressure hydrocephalus (Wet, Wacky, Wobbly) • Liver damage → manganese accumulation basal → ganglia damage • Vascular • Infections (Encephalitis) • Toxins: Mn, CS₂, CO, MPTP (crack cocaine), Methanol

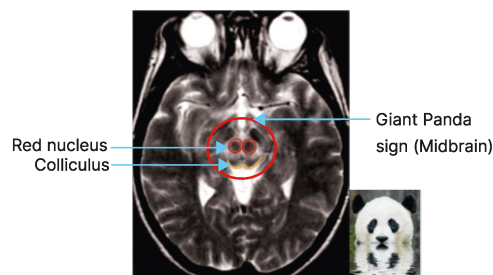
- M/c cause of PD: Sporadic (85%) → genetic/ familial

NEURODEGENERATIVE DISORDERS ASSOCIATED WITH PARKINSONISM

00:44:22

WILSON DISEASE

- Excess of copper
- Autosomal recessive
- ATP 7B gene involved
- Defective hepatobiliary excretion of copper
- Lenticular nucleus involved
- Giant face of panda sign: MRI brain

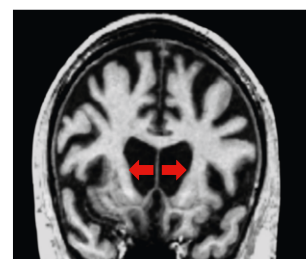


NEURODEGENERATION WITH BRAIN IRON ACCUMULATION

- Halloverden Spatz disease
- Iron accumulation at globus pallidus
- Eye of the tiger appearance: MRI brain

HUNTINGTON DISEASE

- Autosomal dominance inheritance
- Chromosome 4 involved
- Gene: Huntingtin gene involved
- ↑ dopamine/ ↓ GABA
- (+) Chorea + dementia
- MRI: caudate atrophy → appearance of enlarged lateral ventricle
- Bradykinesia, rigidity
- Anticipation: disease occurs in the next generation at an early age

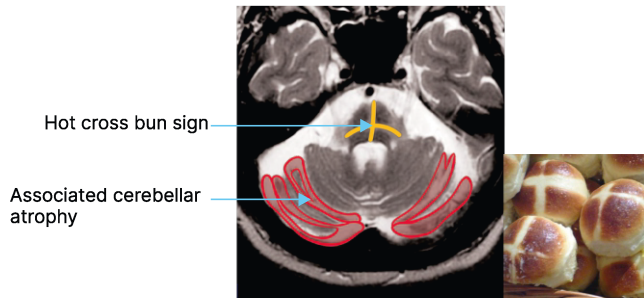


Enlarged lateral ventricle

ATYPICAL PARKINSONISM

MR T2 FLAIR ATROPHY

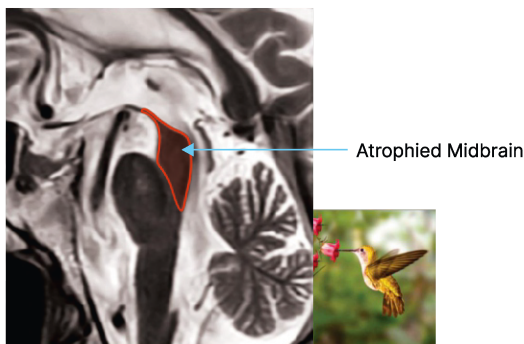
- Rigidity, bradykinesia
- Orthostatic hypotension, neurogenic bladder
- (+) Cerebellar signs (intentional tremors, dysmetria, dysdiadochokinesia)
- MSAP
 - Predominantly Parkinson like manifestations
 - **MRI brain: Iron deposition in striatum**
- MSAC
 - Presence of cerebellar manifestations
 - **MRI: Hot cross bun sign** (associated with cerebellar atrophy)



00:49:24

PROGRESSIVE SUPRANUCLEAR GAZE PALSY

- Tau protein accumulation
- Impaired downward gaze
- Superior colliculus involved
- Recurrent falls
- Retrocollis, neck hyperextension
- Eyelid apraxia (difficulty in opening eyes, absence of blepharospasm) slow ocular saccades
- EOG: square wave jerk
- MRI Brain: **hummingbird appearance**



Hummingbird Sign

CORTICOBASAL SYNDROME

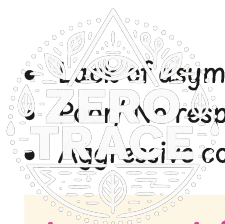
- Tau protein accumulation
- **Alien limb phenomenon**
- Involvement of corpus callosum
- Dementia
- Agnosia (inability to recognize object)

Important Information

- Tauopathies
 - Alzheimer's disease
 - Corticobasal syndrome
 - Progressive supranuclear gaze palsy

POINTERS TO PICK UP ATYPICAL PD IN MCQ:

- Early involvement of gait abnormalities
- Absence of resting tremors (all cases of atypical parkinsonism: PSP, CBS, MSA, FTD)



- Lack of asymmetry
- Poor/No response to levodopa
- Aggressive course

Important Information

- Hot cross bun sign: **MSAc**
- Eye of tiger: **NBIA**
- Hummingbird sign: **PSP**
- Frontal horn width to intercaudate distance ratio ↑: **Huntington disease**

TREATMENT

01:02:37

LEVODOPA + CARBIDOPA (CONTROLLED RELEASE)

- Carbidopa: peripheral decarboxylase inhibitor → ↓ levels of dopamine in blood
- Fixed dose combination
- Inhaled Levodopa: }
- Intestinal gel via implanted jejunal tube } On demand **Off time** patients: symptoms occur as soon the effectiveness of the drug subsides
- S/e:
 - Nausea/ vomiting
 - Orthostatic Hypotension
 - **Punding** (assembly and disassembly of object)
 - Hypersexuality
 - Impulse control disorder (shopping, gambling)
 - Craving for levodopa
- Problems with levodopa:
 - On and Off phenomenon: S/c apomorphine, inhaled levodopa
 - Wearing off: ↑ frequency of medication
 - Peak dose dyskinesias: Choreiform, myoclonic jerk

DOPAMINE AGONISTS

- Acts on D₂ receptors
- Non-ergot > Ergot alkaloids
- Ergot alkaloids: Bromocriptine, Cabergoline
- Non-ergot alkaloids: preferred. Pramipexole, Ropinirole, Rotigotine (transdermal patch)
 - Rotigotine → PD patient undergoing surgery
- S/e: Nausea/ vomiting, hypersexuality, impulse control disorders

MAO-B INHIBITORS

- MOA- blocks central dopamine metabolism
- ↑ Synaptic dopamine availability
- Rasagiline
- Selegiline
- Saffinamide- reversible, anti-dyskinetic



COMT INHIBITORS

- Inhibits the degradation of dopamine
- Opicapone: OD drug, ↑ total duration of action
- Tolcapone: hepatotoxic potential
- Entacapone

ADENOSINE A_{2A} RECEPTOR ANTAGONIST

- MOA- ↓ excessive activation of indirect pathway
- Istradefylline: ↓ off time
- Amantadine
- NMDA antagonist, H₃N₂: influenza A
- ↓ off time
- Zonisamide

DEEP BRAIN STIMULATION: SURGICAL APPROACH

- Indications
 - Advanced PD
 - Peak dose dyskinesia
- Site of electrolyte deployed
 - Subthalamic nucleus
 - Globus pallidus interna
- Results: Tremors ↓



A 60-year-old male presents to the neurology clinic with complaints of tremors, bradykinesia, and stiffness in his limbs. He reports difficulty in initiating movements and experiences tremors primarily in his hands, at rest. On examination, the patient demonstrates cogwheel rigidity and a shuffling gait. What is the likely diagnosis?

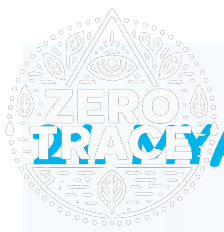
- a. Parkinson's disease
- b. Alzheimer's disease
- c. Corticobasal degeneration syndrome
- d. Myasthenia gravis

Ans: (a)

A 75-year-old banker complains of increasing difficulty buttoning his shirt every morning and brushing his teeth. He states that his hand shakes while relaxing in front of the TV but stops while reaching for the remote and switching channels. Which of the following is the next most appropriate step in managing this patient?

- a. Deep brain stimulation
- b. Send him for cognitive behavior therapy (CBT)
- c. Prescribe L-Dopa
- d. Prescribe Memantine

Ans: (c)



MYASTHENIA GRAVIS



- Autoimmune disorder
- Type II hypersensitivity reaction

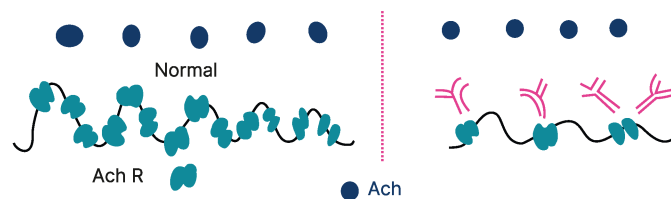
AUTOIMMUNE DAMAGE TO NMJ

00:01:20

- Female: Male ratio → 3:2
- Usual age of presentation
 - Female: 20-30 years
 - Male: 50-60 years
- Defect at the NMJ: **Post-junctional defect**
- Autoantibodies are present due to:
 - **Thymic hyperplasia** (65%)
 - **Thymoma** (10%)

PATHOPHYSIOLOGY

00:02:13



- Decreased acetylcholine receptors at the postsynaptic membrane
 - ACh receptors have 5 subunits
 - ACh → activates α subunit → Na^+ influx → muscle depolarization
 - In MG, less no. of α subunits are present
 - Autoantibody: **Anti-ACh R**
 - Flattening of the postsynaptic folds
 - Presynaptic rundown: acetylcholine release/ impulse ↓
 - Causes:
 - Cardinal feature: Easy fatigability/ weakness
 - Repetitive nerve stimulation: **Decremental response**
- ↓ in the efficiency of neuromuscular transmission

VILLAIN ANTIBODIES

- **Anti-ACh R**: Damages the α subunit of ACh R at the postsynaptic membrane
 - Specific test, not a sensitive test
 - Can be negative also
- **Anti-muscle-specific kinase antibody (Anti-MuSK α/b)**: Prevents clustering of ACh R
 - Associated with poor response to pyridostigmine
- **Anti-LRP-4 antibody** (low-density lipoprotein receptor-related protein 4)
- Antibodies against



Agglin

Agglin 1 receptor

Agglin-2

} Responsible for clustering of ACh receptors

- **Anti-striated muscle antibodies:** Directed against **Titin** (spring protein → gives elasticity to the muscle)

CLINICAL FEATURES

00:10:38

- Easy fatigability and muscle weakness → Cardinal features
- Earliest symptom: **Diplopia** (d/t asymmetrical ptosis)
- **Ptosis**- asymmetrical
 - Triggered by:
 - Sustained upward gaze > 30 seconds (**Simpson test**)
 - Watching TV, games on a mobile phone
 - Diurnal variation of symptoms (normal in the morning, ptosis in the evening)
- **Snarling facies** - Like artificial smile
 - D/t facial muscle weakness
- **Difficulty in chewing**
- **Dysphagia** → Oropharyngeal
- **Dysarthria** and mushy voice → Unclear speech
- Palate weakness: Nasal regurgitation of fluids (Bulbar weakness)
- Proximal asymmetrical limb weakness
 - Shoulder girdle weakness
- **Dyspnea**
 - Diaphragmatic paralysis
- **Normal deep tendon reflexes** + muscle weakness
- Neonatal transient MG
 - Failure to establish respiration
 - Breastfeeding problems in babies
 - IgG: **Anti-ACh R**
 - T/t: **Neostigmine**
 - Good prognosis

WORK-UP

00:21:22

- Ice pack test:
 - Acetylcholinesterase (AChE) ↓ activity with ↓ temperature
 - AChE is responsible for the degradation of ACh
 - Inhibition of ACh degradation → More ACh at NMJ → Longer duration of stimulus → Muscle performance
- Antibody titers:
 - Anti-acetylcholine receptor antibody
 - Most specific test
 - Checked by radioimmunoassay
 - **Positive in 85%** of generalized MG
 - **Positive in 50%** of ocular MG
 - Anti LRP4 antibody
 - Anti-MuSK antibody
- Repetitive nerve stimulation test: Decremental response

- > 40% decrease at the frequency of 3 Hz stimulation
- **Single fiber electromyography** → **Blocking and jitter**

- **Edrophonium test/ Tensilon test:**

- Done in difficult-to-diagnose cases with negative electrodiagnostic testing.
- Can be false positive in ALS
- Edrophonium inhibits breakdown of ACh → longer duration of action of ACh
- Onset of action of Edrophonium > 30 secs
- Sustained upward gaze: > 30-45 secs triggers ptosis (asymmetrical)
 - **Simpsons test:** To induce ptosis by sustained upward gaze
- 2mg of Edrophonium given → ptosis resolves
- Edrophonium can lead to cholinergic crisis (**SLUDGE**) → Antidote: **IV Atropine**
 - **Hypersalivation**
 - **Lacrimation**
 - **Excessive urination**
 - **Diaphoresis**
 - **GI motility** ↑ - diarrhea
 - **Eyes:** Miosis, emesis
 - **Heart:** bradycardia (most reliable indicator), BP ↓
- **Imaging:**
 - **CT chest:** To r/o thymus pathology
 - **MRI Brain:** Ocular MG
- **Autoimmune panel**
 - Hashimoto's thyroiditis: **Anti-TPO antibody**
 - Graves disease: **TSH receptor antibody (TRAb)**
 - RA- **Anti-CCP**
 - SLE- **ANA**
- TB workup
- HbA1c: done to observe the glycemic values
- Bone densitometry
- **Thyroid Function Test**
 - To assess the thyroid
- PFT - done before thymectomy

} Precautionary workup, since the t/t involves long standing steroid administration

TREATMENT

GENERALIZED MG

- **DOC: Pyridostigmine + steroids**
 - When patients develop Cushing's disease symptoms → Mycophenolate instead of Steroids
- **Evaluate for thymectomy**
 - Good FVC (> 15 mm/kg)
 - Remission in most cases
 - Reduces the dose of steroids
- **If symptoms persist:**
 - Steroids - Long-term basis
 - **Mycophenolate**
- **TOC: Thymectomy (Generalized MG)**

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- Symptoms for 3 years (clinical diagnosis)
- T3 ablation
- MRI Brain: to r/o secondary causes
- DOC: **Pyridostigmine (may be worsening)**
 - If symptoms persist, immunotherapy
- Drugs
 - Glucocorticoids
 - Mycophenolate
 - Rituximab
 - Eculizumab

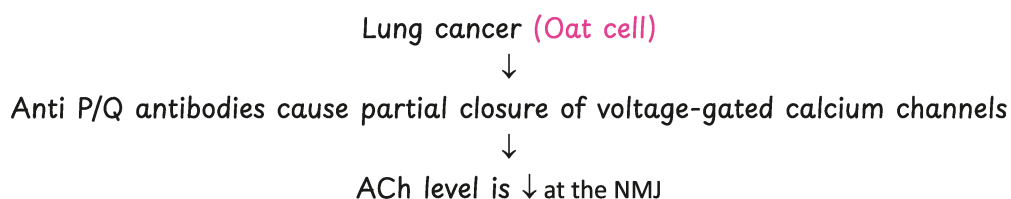
MYASTHENIC CRISIS

Important Information

- IVIG used in the t/t of:
 - GBS
 - Kawasaki disease
 - Myasthenic crisis
- Diaphragmatic paralysis
 - Requirement of ventilatory support
- Example of Type II respiratory failure
- Trigger: Intercurrent infections (COVID-19, UTI, Sepsis)
- Drugs
 - **DOC: IV immunoglobulins (IVIG)**
 - Plasmapheresis

LAMBERT EATON SYNDROME

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- **Presynaptic disorder**

CLINICAL FEATURES

- A 50-60-year-old male smoker presents with:
 - Proximal limb weakness
 - Ptosis
 - Diplopia
 - DTR: ↓
 - Dry mouth
 - Impotence